



Capstone Project

Car Resale Value Analysis

Group **G10**

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Introduction

Why This Matters



Growing Pre-Owned Vehicle Market

Expanding rapidly with significant opportunities



Key Decision-Makers

Buyers, sellers, and dealers rely on accurate pricing



Lack of Transparent Pricing

No standardized valuation mechanisms exist



Financial Impact

Critical influence on automotive industry economics

Problem Statement

What Gap Are We Addressing?



Inconsistent Pricing

Wide variation in vehicle valuations across different platforms



No Standardized Models

Lack of data-driven, objective valuation frameworks



Subjective Dealership Pricing

Dealer valuations often lack transparency and consistency



Information Asymmetry

Unequal knowledge between buyers and sellers

Objective

Main Aim

Develop a **data-driven pricing model** for used cars

Measurable Goals



Identify key **pricing factors** impacting resale value



Build predictive model with **R^2 and RMSE validation**



Provide actionable pricing recommendations

Dataset Overview

Comprehensive Market Data



Total Records: 11,995 Vehicles

Comprehensive dataset spanning 25 years of market data



17 Structured Features

Including price, brand, year, mileage, fuel type, and more



Time Period: 1995 – 2020

Price Range: ₹25,000 to ₹60 Lakhs



Fuel Types: Petrol, Diesel, CNG, LPG, Electric

Transmission: Manual & Automatic

Methodology

Research Process



Data Collection

Gather raw dataset from market platform



Data Cleaning

Handle missing values, remove outliers



Exploratory Data Analysis

Discover patterns and correlations



Advanced Analysis

Segmentation and feature evaluation



Insights & Recommendations

Generate actionable findings

Statistical Techniques Used



Correlation Analysis

Identify relationships between variables



Outlier Detection

IQR Method for data quality



Feature Importance

Rank pricing factors by impact



Regression Modeling

Build predictive pricing model



R^2 and RMSE metrics for model validation

Key Observations



Petrol Vehicles Dominate Dataset Share

Highest market presence among fuel types



SUVs Show Higher Average Resale Price

Premium segment compared to hatchbacks



Newer Vehicles Retain Higher Resale Value

Post-2015 models show better value retention



Strong Negative Correlation: Mileage vs Price

Higher kilometers driven reduces resale value

Key Findings

Important Patterns Discovered



Vehicle Age Shows Strong Inverse Relationship

Older vehicles demonstrate consistent value depreciation patterns



Kilometers Driven Negatively Impacts Value

Higher usage intensity correlates with lower resale prices



Brand Significantly Influences Pricing Variance

Brand reputation affects resale value across market segments



First-Owner Vehicles Command Higher Value

Fewer owners increases buyer confidence and pricing

Business Implications

Data-Driven Applications for Stakeholders



Dealers

Reduce inventory aging using **depreciation trend analysis**. Optimize pricing strategies based on statistical models to minimize holding time and maximize turnover.



Buyers

Benchmark fair value using **statistical pricing range** derived from 11,995 vehicle records. Make informed purchase decisions with transparent valuation.



Insurers

Reduce claim disputes using **data-driven valuation models**. Standardized pricing based on statistical analysis enhances trust and reduces conflicts.



Platforms

Build market trust through **transparent pricing models**. Data-backed insights attract users seeking reliable, unbiased vehicle valuations.

Challenges Faced

Data & Analysis Limitations



Missing Values

Mileage & engine power columns contained null values requiring imputation



High Price Variance

₹25K to ₹60L range created wide distribution requiring careful analysis



Luxury Vehicle Outliers

Premium segment vehicles skewed average pricing statistics



Categorical Standardization

Feature encoding required for consistent model training

Conclusion

Key Pricing Drivers Identified

Analysis based on **11,995 vehicle records** provides strong statistical foundation

1

Vehicle Age

Primary driver

2

Kilometers

Usage metric

3

Brand

Value factor

4

Owner Type

Trust indicator

5

Fuel Type

Preference driver



Objectives Achieved

Data-driven pricing model successfully developed



Actionable Insights

Clear recommendations for all stakeholders

Future Scope & Thank You

Future Enhancements



Expand Dataset Beyond 2020

Include latest market data for improved accuracy



Real-Time ML Pricing Model

Deploy dynamic pricing system for instant valuations



Web-Based Valuation Tool

Build user-friendly interface for vehicle pricing



Dealer System Integration

Connect with automotive platform APIs



Thank You

We appreciate your time and attention

Group G10

Car Resale Value Analysis Team